This Dynamic Planet
Plate Tectonics and Earthquakes

Overview
This is a two-part lesson. The first part simplifies the idea of plate tectonics and might be best used with elementary or middle school students. The activity uses familiar objects to teach unfamiliar concepts: the cross section of an apple to illustrate the earth's structure and a soccer ball to demonstrate earth's interlocking plates. With this background in mind, students use Encarta Encyclopedia to apply the idea of plate tectonics to the movement of continents, particularly the breakup of Pangaea. The second part of the lesson links plate tectonics to earthquakes through use of maps found on Encarta® Interactive World Atlas. In the World Atlas, students use the Map Customizer to locate earthquake belts and sites of major earthquakes in the past. They then make predictions about the location of future earthquakes.
This lesson can be used with a related lesson on plate tectonics and volcanoes, The Ring of Fire.

Objectives
In completing this lesson, students:
1. Describe the theory of plate tectonics
2. Use plate tectonics to explain continental drift
3. Discuss the link between plate tectonics and earthquakes
4. Predict the regions most likely to experience earthquakes in the future

Materials
Part One: Globe, soccer ball, apple, knife (to cut the apple)
Part Two: Tectonic map showing plates and earthquakes (can use Encarta Interactive World Atlas)
Parts One and Two: Encarta Encyclopedia 2000

Background Information/Fast Facts
In introducing the activity, you might want to share some of the following facts with students. (All information comes from Encarta Encyclopedia.)
- Earthquakes, or seismic tremors, occur at a rate of several hundred per day around the world.
- Approximately one million small earthquakes shake the world each year.
- Nearly 80 percent of seismic energy results from tremors along the Pacific Rim.
- Japan is shaken by more than 1,000 tremors each year.
- The largest earthquakes in the United States have taken place in the Alaskan islands.
- Scientists have discovered extraterrestrial quakes on other planets and their satellites.
- Seismic events on the moon are known as moonquakes.

Activities and Procedures
PART ONE
1. Place a globe, a soccer ball, and an apple in front of the classroom. Ask students if they can guess what all three of these items have in common. (Students will probably focus on the shape of the objects.) After a few minutes, explain that each object represents a different model of earth.

2. Cut the apple in half lengthwise and hold the cross section up to the class. Tell students that the cross section represents the basic structure of earth: a thin outer crust (the apple's skin), a small inner core, and a vast molten interior.
3. Now hold up the soccer ball. Point out that the covering appears to be made of many pieces. Explain that scientists believe that in some ways this is how the earth's crust looks—a collection of pieces that fit together in different ways. Unlike a soccer ball, however, the pieces of crust that cover the earth move. In fact, they move all the time. Tell students that scientists call these chunks of crust "plates" and refer to their movements as "plate tectonics."
4. Next, focus student attention on the globe. Mention that scientists think that millions of years ago there was only one huge continent on earth. Challenge students to use what they know about the earth's structure and crust to speculate on why there are so many continents today. Students will most likely say "the pieces of crust moved around." Use this or similar responses to raise the central issue of how and why the plates move.
5. To explore this issue, write the following question on the chalkboard: "How and why do continents shift and rift and drift apart?" Assign students to use Encarta to answer this question. You might want to assign students particular tasks. For example, one group might check Encarta Encyclopedia articles; another group might investigate articles included in Geography in Depth in Encarta Interactive World Atlas; yet another group might focus on the World Atlas Map Treks.
6. When students have finished their research, organize discussion in the form of a geological conference set in the late 1960s or early 1970s. You, or a student volunteer, can set the mood of the conference by using "This Changing Earth," a 1973 National Geographic story found in the Encarta Encyclopedia Article Archives.

PART TWO

1. Request a volunteer to use the Encarta Interactive World Atlas to locate and print a map showing tectonic plates and earthquake belts around the world. Make enough copies for all the students in your class.
2. Introduce the subject of earthquakes by sharing the Fast Facts. Ask students what they can infer about earthquakes based on these facts. *(That earthquakes occur frequently throughout the world, particularly along the Pacific Rim, and in other parts of the solar system.)*
3. Next, distribute copies of the tectonic map. Ask students to discuss reasons earthquakes might occur along the tectonic plates. Have students check their answers by reading articles on earthquakes found in the encyclopedia and the Geography in Depth feature.
4. Review the connection between tectonic activity and earthquakes, focusing on subduction zones and the elastic rebound theory. Ask students to indicate other causes of earthquakes (e.g., volcanic eruptions, human activities).
5. Remind students that one of the ways seismologists predict future earthquakes is by studying the location and severity of past earthquakes. Tell students to use Encarta's reference tools and multimedia to find tables listing the major earthquakes of the past, particularly those occurring during the 20th century.
6. After students have located this information, direct them to locate these sites, along with dates and magnitude, on a tectonic map of earthquake belts. Looking at the completed maps, ask students to indicate future danger zones. Elicit evidence in support of their predictions.
7. In a follow-up discussion, encourage students to discuss emergency preparedness plans. Do not neglect related dangers such as tsunamis. At this

point, you might also want to incorporate one or more of the Alternate Activities into the lesson.

Alternate Activities

Designing an Earthquake-Resistant Home

The articles in Encarta Encyclopedia include a number of film clips and photographs showing earthquake destruction, particularly in Mexico City and Kobe, Japan. Tell students to imagine they are part of a team of architects and engineers hired to design earthquake-resistant homes for the people of Mexico City or Kobe. Students should begin their study by viewing the film clips and photos. They should also read the suggestions in the encyclopedia article **Earthquake** and the **Geography in Depth** article in Encarta Interactive World Atlas. (Remind students to follow the link to the related topic **Civil Engineering** in the Encarta Encyclopedia **Earthquake** article.) When students are done, have them create an architectural plan in the form of a drawing or a blueprint. If possible, you might invite a local architect or civil engineer to review the plan for structural soundness. Encourage students to make the necessary modifications in their drafts.

Creating an Earthquake Survival Guide

Assign interested students to search the Web for pamphlets and booklets on earthquake preparation. (Some possible sites are suggested at the end of this lesson.) Based on these materials and information in Encarta, have students create an earthquake survival guide for people their own age. You might arrange for students to share their guides with schools in earthquake-prone areas. If students live in an area that has experienced earthquakes in the past, you might arrange for the guides—along with pertinent reference materials—to be placed on reserve in the school or community library.

Linking Earth Science and Life Science

Can animals predict earthquakes? Some people think so. Before the 1974 Haicheng and 1975 Tangshan quakes, pandas reportedly held their heads and screamed, horses stampeded and neighed, and snakes emerged from their underground dens in the dead of winter. Recount this information to students and have them research evidence of animal seismologists. In addition to libraries and Web sites, students might talk with experts in animal behavior and, if possible, interview people who have lived through earthquakes. The key question is: Did you observe any unusual animal behavior in the hours, days, or weeks before the quake?

Writing an Earthquake FAQ Sheet

Call on volunteers to explain the meaning of the acronym FAQ. (If your students don't know the answer to this question, refer them to Encarta's Computer and Internet Dictionary or remind them that the purpose of a FAQ Sheet is to answer the most common questions about a subject, hence the name "**F**requently **A**sked **Q**uestions.") Now assign students, individually or in groups, to draw up a FAQ Sheet about earthquakes. Consider the possibility of submitting their finished FAQ Sheet to one of the educational Web sites for kids.

For articles, maps, or other media pertaining to this topic, look up the following in Encarta® Encyclopedia:

Continent
Earthquake
Geology
Mountain
Pangaea
Plate Tectonics
Seismology
Tsunami

Volcano
Continental Drift Animation

Refer to the following items in Encarta® Interactive World Atlas for relevant information:
Continental Drift, Glossary
Plate Tectonics, Global Theme

For additional online information and media, view the following Encarta articles:
Earthquake
Plate Tectonics

For additional relevant information, visit the following Web sites:
Plate Tectonics, the Cause of Earthquakes
http://www.seismo.unr.edu/ftp/pub/louie/class/100/plate-tecton
ics.html
The Nevada Seismological Laboratory uses aerial pictures and diagrams to show how the movement of the earth's crust can cause deformations and earthquakes.
The University of Nevada, Reno Seismological Laboratory
http://www.seismo.unr.edu/index.html
The Nevada Seismological Laboratory offers a wide variety of earthquake-related information and links to related sites.